

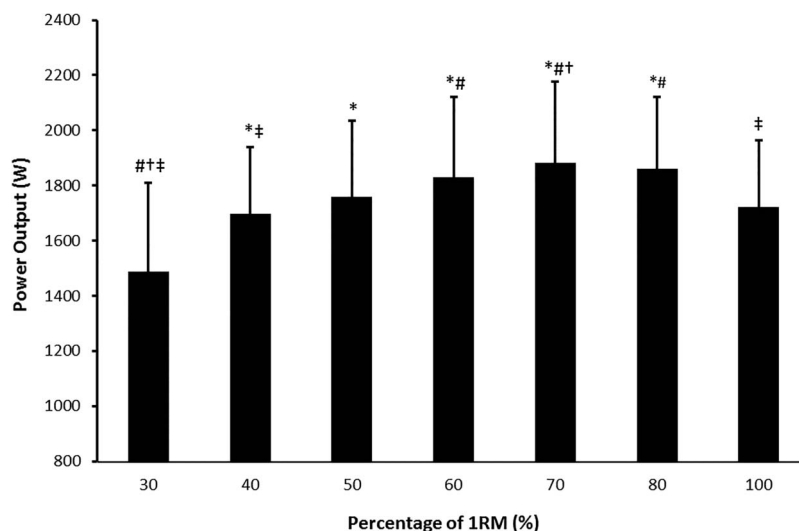


Figure 1. Mean $\pm$ SD power output (W) across a range of 1 repetition maximum (1RM) percentages during the high pull. *Denotes significant difference to 30% 1RM, #denotes significance difference to 40%, ‡denotes significant difference to 50%, and †denotes significant difference to 80%; all $p < 0.05$.

patterns, due to excessive loading or poor technique, may contribute to an increased risk of injury. Using electromyography (EMG), muscle activation patterns and the onset of muscle activation have been assessed in the lower body (19,20,25). For optimal force development and movement control, activation in muscles crossing a common joint is likely to occur in a certain, coordinated order. However, little is known about the sequencing of muscle activation during dynamic, multijoint exercises, and therefore, it is currently unclear whether normal activation patterns during the Olympic lifts, and their derivatives, involve asynchronous or synchronous onset of muscle activation. Understanding onset patterns across a range of loads may improve technique development and may assist to identify injury potential. Recently, Cochrane and Barnes (4) investigated 4 submaximal loads in the deadlift and reported synchronous activation of the GM and BF, irrespective of load. This finding is in contrast to the results of others investigating activation of the hip extensors during isolated hip extension in the prone position (2,19,33).

Using an Olympic lift variation that does not include the catch phase, such as the HP, is regarded as a less technical exercise than the full Olympic lift that promotes lower-body muscular power (29). This may reduce the risk of upper-limb injury while generating high power outputs because of triple extension. Therefore, the 2 objectives of this study was, first, to determine at what percentage of 1RM PPO occurs during the HP, and second, to determine the onset time of the hip extensors in men when performing the HP across a range of loads. Given the findings of previous, comparable studies, we hypothesized that PPO will occur at a heavy load (60–80%

of 1RM), and that onset of muscle activation during hip extension will be similar irrespective of loading.

METHODS

Experimental Approach to the Problem

At least 24 hours before testing, subjects attended the laboratory for familiarization with the procedures used in the study and to have their HP technique assessed by a suitably qualified trainer; if technique was deemed inadequate, subjects were excluded from the study. On another occasion, subjects reported to the laboratory at a convenient time for testing, having abstained from exercise in the previous 24 hours. Subjects completed 1RM HP testing followed by a 10-minute rest. Subjects then performed HP at 30–80%, and 90% if required, of their recently established 1RM to ascertain PPO.

Subjects

Twelve healthy, resistance-trained men (mean $\pm$ SD: age 24.8 ± 5.8 years, 19–38 years; height 1.80 ± 0.1 m; body mass 96.4 ± 17.3 kg) volunteered for the study. All subjects had at least 2-year resistance training experience and were training at least 3 times per week leading into the study. Subjects' HP 1RM was 91.3 ± 15.3 kg. Ethical approval was granted by Massey University Human Ethics Committee, and written informed consent was obtained from subjects.

Procedures

One Repetition Maximum and Peak Power Output Testing. One repetition maximum testing was executed in accordance

TABLE 1. Mean $\pm$ SD onset time and relative onset of the biceps femoris (BF) gluteus maximus (GM) during the high pull at a range of percentage 1 repetition maximum (1RM).

	Percentage of 1RM (%)						
	30	40	50	60	70	80	100
Onset time BF (ms)	212.1 ± 225.9	163.8 ± 73.3	156.5 ± 97.8	163.5 ± 97.8	176.8 ± 62.4	191.1 ± 78.2	158.5 ± 88.6
Onset time GM (ms)	130.4 ± 59.6	112.1 ± 25.7	148.8 ± 103.3	139.0 ± 50.5	158.1 ± 80.5	141.4 ± 55.7	141.5 ± 46.3
Relative (BF-GM) onset (ms)	81.7	51.7	7.7	24.4	18.8	49.7	17.0

with Brown and Weir (1) with load increased incrementally, depending on successful completion of the previous attempt, with 2- to 5-minute rest between each maximal attempt. Starting with the loaded Olympic weightlifting barbell on the ground, subjects were required to pull the bar upward in a single, continuous effort, so that full extension occurred at the trunk, hips, knees, and ankles, and so, the bar reached nipple line (top of the pull). If these criteria were not attained, the lift was deemed unsuccessful after an appropriate rest; another attempt was performed.

Once the subjects 1RM was determined, a 10-minute rest was given before subjects completed 2 repetitions at 10% increments from 30 to 80% of 1RM. For ease and subject safety, the loading order was not randomized. To ensure that PPO was accurately established, efforts at 90% of 1RM were made if PPO occurred at 80%. Repetitions were performed consecutively at each load with subjects instructed to re-set after the first rep to avoid bouncing the barbell off the ground. Subjects were instructed to move the barbell as fast as possible during all efforts. Each load was interspersed by 2- to 5-minute rest, so maximal effort could be attained. The highest PPO from the 2 repetitions at each load was used for subsequent analysis.

Power output was measured by a custom-made, fine wire, linear potentiometer (0.03-mm resolution; Jinan, Kaisi Technology Co Ltd., Shandong, China) attached to the middle of the barbell. This provided displacement, which was then converted into power output in Chart for Windows v7 (ADInstruments, Dundee, New Zealand). From our laboratory, the coefficient of variation of the fine wire potentiometer was calculated at 5.2%. Instantaneous velocity and acceleration was calculated from first- and second-order derivatives, and force was the product of mass of bar + load multiplied by bar acceleration + gravitational acceleration. Power was calculated from the product of force and velocity, and peak power was determined as the highest power output over the lifting phase. Data were sampled at 1,000 Hz and filtered with a low-pass filter of 5 Hz.

Electromyography. Muscle activity of the left BF and GM muscles was measured using surface EMG. Before electrode placement, the site was measured, shaved, abraded, and cleaned with isopropyl alcohol. Highly conductive wet gel Ag-AgCl sensors (22 mm diameter; Ambu A/S, Ballerup, Denmark) were placed on the midbelly of the muscle parallel to the direction of the muscle fibers in accordance with Hermens et al. (10). Wireless EMG units were secured by adhesive tape and connected to the electrodes, and the EMG signals were transmitted to a personal computer interface receiver (Telemyo DTS; Noraxon, Scottsdale, AZ, USA). The EMG signals were processed by a data acquisition system (MyoResearch XP; Noraxon) that was synchronized with a video camera (HD Webcam C615; Logitech, Hong Kong, China). The EMG data were sampled at 3,000 Hz, filtered with a low-pass filter of 500 Hz and rectified and

smoothed over 100 milliseconds (root mean square algorithm). Timing of onset was defined as the point where EMG activity exceeded the peak resting amplitude by 20–50% for at least 50 milliseconds (23). The relative onset time of BF and GM (BF-GM) was calculated according to Chance-Larson et al. (3), where GM onset was subtracted from BF onset times (ms). A positive time value indicated that GM was activated before BF, whereas a negative time value indicated that BF was activated before GM.

Statistical Analyses

A general linear model repeated-measures analysis of variance was used to evaluate peak power and onset times of BF and GM. When a significant main effect was attained, post hoc pairwise comparisons were performed using the Bonferroni procedure. Data were analyzed using the IBM SPSS Statistics (Version 22.0, IBM Corp, NY, USA) values reported as mean $\pm$ SD. The statistical significance was set at a level of $p < 0.05$, and when significance was acquired, the effect size (ES) was calculated. Effect size magnitudes were classified as trivial, 0–0.2; small, 0.2–0.6; moderate, 0.6–1.2; large, 1.2–2.0; and very large, >2.0 (12).

RESULTS

Peak power output during the HP occurred at 70% of 1RM (1881.9 ± 296.1 W), although this power output was not different to 60% (1829.3 ± 292.6 W, trivial ES = 0.18), 80% (1862.4 ± 258.4 W, trivial ES = 0.07), and 100% (1722.6 ± 240.8 W, small ES = 0.56) of 1RM (91.3 ± 15.3 kg) (Figure 1). Power output was significantly lower (moderate to large ES = 0.73–1.28; $p < 0.05$) at 30% of 1RM ($1,489.8 \pm 321.6$ W) compared with all loads, except for 100% 1RM (moderate ES = 0.82; $p = 0.082$). Furthermore, power production at 80% of 1RM was significantly greater ($p < 0.05$) than at 30% (large ES = 1.28), 40% (moderate ES = 0.66), and 100% (small ES = 0.56) of 1RM (Figure 1). Peak power output at 70% of 1RM occurred in 6 of the 12 subjects; the remaining 6 subjects recorded their PPO at 80% of 1RM.

For onset time, no significant difference ($p > 0.05$) occurred between the various loads and muscles (BF and GM), and there was no load $\times$ muscle interaction effect ($p > 0.05$) (Table 1). Furthermore, no significant difference ($p > 0.05$) occurred in relative onset (BF-GM) between loads (Table 1).

DISCUSSION

Our hypothesis that a higher 1RM percentage (60–80% of 1RM) would elicit a significantly higher power output, compared with lower loads, based on previous research using Olympic lifts (5,7,18,24,30,31,35), was confirmed. Although PPO occurred at 70% of 1RM, this was not significantly different to power outputs at 60 and 80% of 1RM suggesting that, for training purposes, a range of loads above 60% may be appropriate to enhance lower-body PPO. No significant difference was noted in the onset of muscle activation between loads or muscles.

There is limited research investigating PPO during the HP; 2 studies examined the hang pull; however, only a limited number of loads were included (30,31). This study is the first, to our knowledge, to investigate where PPO occurs over a large number ($n = 7$) of 1RM percentages using this movement. The results of this study are similar to other studies in which Olympic lifts, and their variations, produced PPO at 70 and 80% of 1RM (5,15,18,24,35). Consequently, and similar to our findings, some of these studies also reported no statistical difference across a number of relative loads (15,18). Cormie et al. (5) investigated different 1RM% in the hang power clean in professional rugby players with PPO at 80% of 1RM, and no statistical difference observed between 50 and 90% 1RM. In the current study, the number of loads investigated and the small, 10% 1RM, increment between these loads, resulted in a loading window (60–100% 1RM) where no significant difference in power output occurred; this is not unexpected as small changes in load are likely to have only small effects on bar velocity and therefore power output.

Across all 1RM%, the onset of GM (138.8 ± 60.2 ms) activation occurred before BF (174.6 ± 97.3 ms), although no significant differences occurred between the 2 muscles or across the 7 loads. Consequently, there was no effect of increasing HP load (30–100%) on relative BF and GM onset time. The relative onset time (BF-GM) of the current study was similar to those reported previously (4,14,19). However, 2 of those studies were conducted using body weight prone hip extension (14,19), whereas the other study, Cochrane and Barnes (4), used the deadlift which is similar to HP, as both movements involve triple extension (ankle, knee, and hip) and are closed kinetic chain exercises. Accordingly, from the literature, there are conflicting results with GM being the last muscle to be activated during the prone hip extension (2,19,25,33), whereas in this study and in Cochrane and Barnes (4), GM was not the last muscle to be recruited. The differences in onset during open- and closed-chain movements may explain the divergent results. Investigating the onset of muscle activation of the quadriceps during open and closed kinetic chain exercises, Stensdotter et al. (27) found the onset of EMG activity of the quadriceps (all 4 muscles) to be more synchronous during a closed-chain exercise than in an open-chain exercise. In addition, Mellor and Hodges (22) reported greater synchronization of motor units between the vastus lateralis and vastus medialis obliquus during closed-chain compared with open-chain tasks. These findings, together with those of Cochrane and Barnes (4) and our present results, suggest that, during closed chain movements, muscle groups acting across a common joint are likely to act in a relatively synchronous manner that ensures optimal force production and control of movement. Further research using a range of loads is required to confirm this theory.

PRACTICAL APPLICATIONS

Peak power output occurred at 70% of 1RM, although there was no significant difference between 1RM% at 60–80%

(80% was significantly greater than 100%). Therefore, these findings suggest that training in these ranges is likely to maximize power output. Although the GM is activated before the BF, albeit nonsignificantly, irrespective of the percentage of 1RM, there was no significant effect of increasing load on relative BF and GM onset time. Practitioners can be confident that if HP technique is enforced over a range of loads, particularly heavy loads, it will ensure consistent muscle sequencing to developing force across the hips. This may prevent a possible compromise delay in GM, which has been associated with lower extremity injuries.

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